

BIOS 735: Bagging and Random Forests

Alexander McLain

September 22, 2026



Why Go Beyond CART?

- ▶ CART trees are simple and easy to interpret.
- ▶ However, they can be **unstable** and produce less accurate predictions.
- ▶ **Goal:** build a method that is more **stable and accurate**.
- ▶ **Solution:** combine many trees to smooth out variability.

Bagging

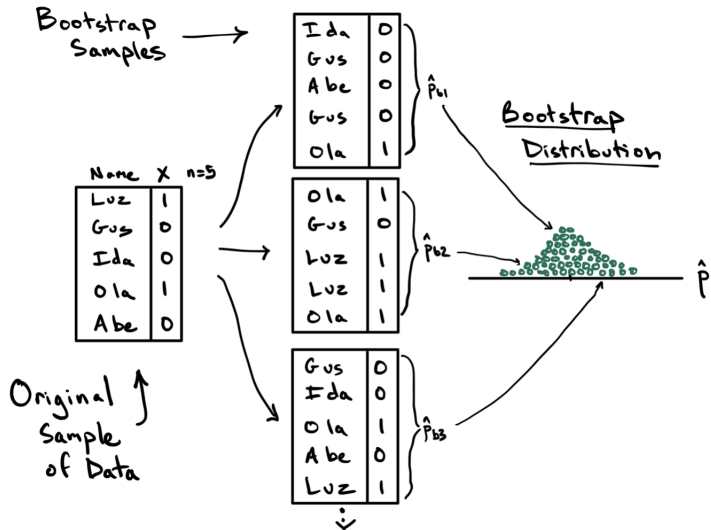
- ▶ Bagging and Boosting are machine learning methods that can be used to improve a model.
- ▶ Bagging or *bootstrap aggregation* is a technique for reducing the variance of an estimated prediction function.
- ▶ Bagging is particularly effective for predictions that have low bias and high variance.
- ▶ Today, we'll focus on bagging CART.

Bootstrapping

- ▶ Bootstrapping uses the observed data as an approximation to the population.
- ▶ Given n observations $\mathcal{D} = \{(y_1, \mathbf{x}_1), (y_2, \mathbf{x}_2), \dots, (y_n, \mathbf{x}_n)\}$ sample with replacement to get $\mathcal{D}^{(b)}$ that has n observations.
- ▶ Each bootstrap sample ($\mathcal{D}^{(b)}$):
 - ▶ contains n observations;
 - ▶ includes some observations multiple times;
 - ▶ leaves some observations out.
- ▶ Calculate the statistic of interest (θ) for each bootstrap sample:

$$\hat{\theta}^{*(1)}, \hat{\theta}^{*(2)}, \dots, \hat{\theta}^{*(B)}.$$

- ▶ The distribution of the bootstrap estimates approximates the **sampling distribution** of the statistic.



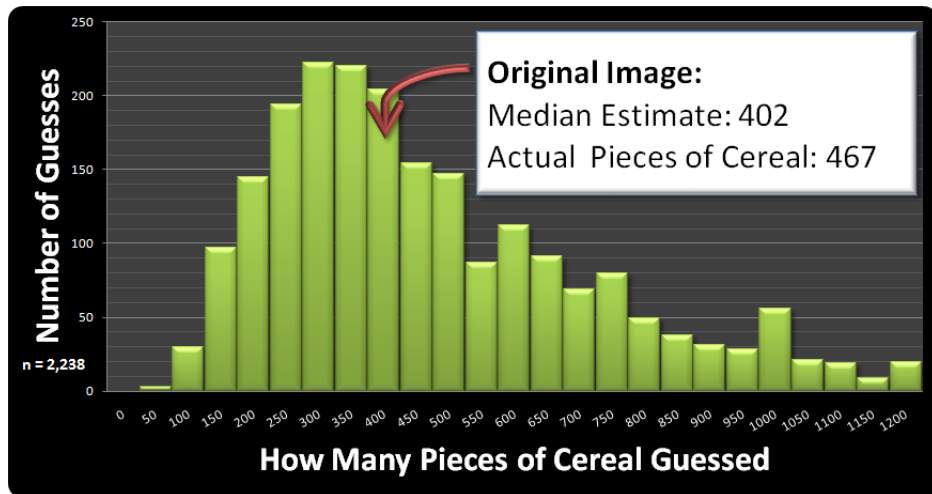
The Idea of Bagging

- ▶ **Bootstrap Aggregating (Bagging):**
 1. Take many bootstrap samples from the data.
 2. Fit a CART model to each sample.
 3. Average predictions (for continuous outcomes) or use majority vote (for classification).
- ▶ Bagging reduces variance and improves prediction accuracy.
- ▶ Example: Predicting hospital readmission rates.
 - ▶ One tree may split first on age.
 - ▶ Another may split first on the length of stay.
 - ▶ Another on comorbidities.
- ▶ Bagging combines them for a more stable prediction.

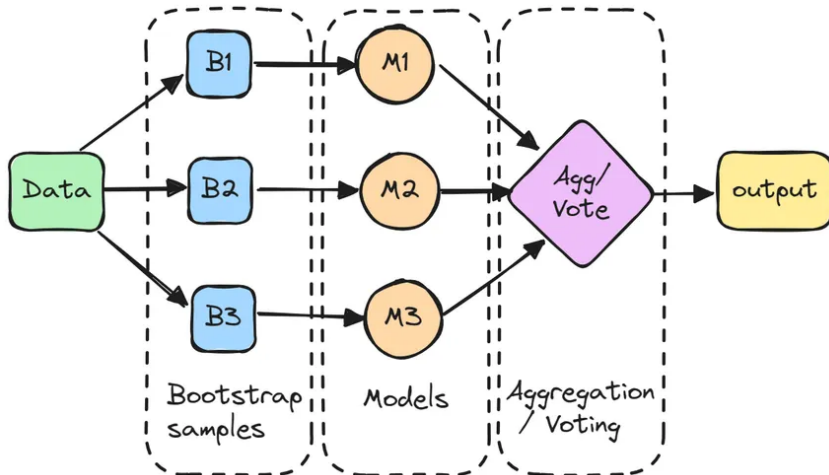
Wisdom of crowds

Sir Francis Galton's Ox Experiment (1906):

- ▶ Sir Francis Galton, a British polymath and cousin of Charles Darwin, was intrigued by the idea of average judgment and wanted to test its validity. He found his opportunity at a country fair in Plymouth, where an event was being held in which attendees could guess the weight of an ox on display.
- ▶ About 800 people, both experts (butchers and farmers) and non-experts alike, took part in the contest, writing down their guesses on tickets.
- ▶ After the event, Galton collected and analyzed all the tickets. He wanted to prove that the average guess of the crowd would be way off the mark.
- ▶ To his astonishment, the mean of all the guesses (1,197 pounds) was incredibly close to the actual weight of the ox, which was 1,198 pounds!



Bagging Illustration



Bagging vs. Boosting

- ▶ **Bagging:** fit many models **independently** and combine their predictions.
- ▶ **Boosting:** fit models **sequentially**, with each new model focusing on observations that previous models predicted poorly.
- ▶ Bagging primarily reduces **variance**.
- ▶ Boosting primarily reduces **bias**, although it can also affect variance.

Random Forests

- ▶ *Random forests* (Breiman, 2001) is a modification of bagging CART that builds a large collection of de-correlated trees, and then averages them.
- ▶ *Random forests* are very simple to train and use.
- ▶ Bagging is useful when we have many noisy but unbiased classifiers.
- ▶ CART are very noisy and unbiased (if they are grown deep enough).

Random Forests

- ▶ If variables are identically distributed with correlation ρ then the variance of their average over B samples is

$$\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

- ▶ Which can be made as small as necessary if $\rho = 0$.
- ▶ The idea in random forests: **is to improve the variance reduction of bagging by reducing the correlation between the trees, without increasing the variance too much.**

Random Forests Algorithm

1. For $b = 1, 2, \dots, B$
 - 1.1 Draw a bootstrap sample $\mathbf{Z}^*$ of size N from the training data.
 - 1.2 Grow a random-forest tree T_b to the bootstrapped data, by recursively repeating the following steps for each terminal node of the tree, until the minimum node size n_{min} is reached.
 - 1.2.1 Select m variables at random from the p variables.
 - 1.2.2 Pick the best variable/split-point among the m .
 - 1.2.3 Split the node into two daughter nodes.
2. Output the ensemble of trees $\{T_b\}_1^B$.

Random Forests

- ▶ The algorithm results in $\{T_b\}_1^B$, which can be used to predict classes or continuous outcomes.
- ▶ **Regression:** $\hat{f}_{\text{rf}}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$
- ▶ **Classification:** Let $\hat{C}_b(x)$ be the majority vote for the b th tree, then

$$\hat{C}_{\text{rf}}^B(x) = \text{majority vote } \left\{ \hat{C}_b(x) \right\}_1^B$$

Random Forests Details

- ▶ Typically, values for m depend on whether regression or classification is being done.
- ▶ For classification, the default value for m is $\lfloor \sqrt{p} \rfloor$ and the minimum node size is one (also 1 is not a crazy value).
- ▶ For regression, the default value for m is $\lfloor p/3 \rfloor$ and the minimum node size is five.
- ▶ In practice, the best values for these parameters will depend on the problem, and they should be treated as tuning parameters (**but they are the only tuning parameters**).

Out-of-bag Samples

- ▶ Note that for each bootstrap sample, there are observations ($\sim 37\%$) that are not included in the training of the tree. We call these *out-of-bag* (OOB) samples.
- ▶ The OOB samples can be used to estimate the error.
- ▶ The error for observation $z_i = (x_i, y_i)$ is defined as its prediction error for a random forest, averaging only those trees corresponding to bootstrap samples in which z_i did not appear.
- ▶ Random forests have built-in validation!!

Variable Importance

- ▶ There are two methods to construct variable importance plots for random forests.
- ▶ The first *Gini* approach is described as follows:
 - ▶ At each split in each tree, the improvement in the split-criterion is the importance measure attributed to the splitting variable. It is accumulated over all the trees in the forest separately for each variable.
 - ▶ The improvement in the split-criterion is measured using the *Gini* index.

Variable Importance

The second variable importance measure uses the OOB samples:

- ▶ When the b th tree is grown, the OOB samples are passed down the tree, and the prediction accuracy is recorded.
- ▶ Then the values for the j th predictor variable are randomly permuted in the OOB samples, and the accuracy is again computed.
- ▶ The decrease in accuracy as a result of this permuting is averaged over all trees, and is used as a measure of the importance of variable j in the random forest.

Overview: advantages

- ▶ **Accuracy:** Random forests often provide higher accuracy in comparison to single decision trees.
- ▶ **Overfitting:** The model is less prone to overfitting than individual decision trees, thanks to averaging out the results.
- ▶ **Handling Large Data:** Can efficiently process large datasets and handle datasets with higher dimensionality.
- ▶ **Missing Data:** Can handle missing data by using the median to replace continuous variables or computing the weighted average of missing values.
- ▶ **Out-of-Bag (OOB) Score:** The OOB sample can be used as a validation set to estimate accuracy, removing the need for a separate validation set.
- ▶ **Variable Importance:** Provides insights into feature importance, which can be crucial for domain understanding and feature selection.

Overview: disadvantages

- ▶ **Complexity and size:** The model can be quite large and not as easily exportable and slow for real-time predictions.
- ▶ **Interpretability:** Random forests are more complex than single decision trees and are hence less interpretable.
- ▶ **Bias:** If the dataset has a dominant class, random forests can be biased towards that class. Balancing the dataset prior can help alleviate this.
- ▶ **Parameter Tuning:** They require fewer parameter tunings than some other algorithms, but they still have hyperparameters.
- ▶ **Not Always Best:** For data that can be linearly separated, simpler models like logistic regression might perform better.

CART vs. Bagging vs. Random Forests

	CART	Bagging	Random Forests
Stability	Low	Higher	Higher
Accuracy	Moderate	Higher	Highest
Interpretability	High	Medium	Lower
Notes	Single tree	Averages many trees	Adds predictor randomness